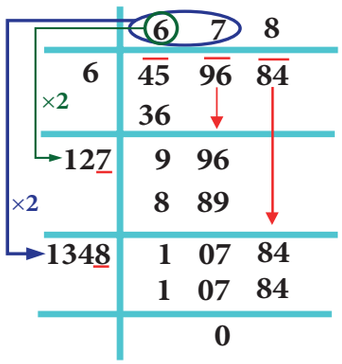
Example 1.27

Find the square root of 459684 by long division method.

Solution:

By long division method, we can find the square root of 459684 as given below:



$\therefore \sqrt{459684} = 678$

Try these

Find the square root by long division method.

1. 400 2. 1764 3. 9801

1.7.3 Number of digits in the square root of a perfect square number

We made use of bars to find the square root in the division method. This marking bars help us to find the number of digits in the square root of a perfect square number. Observe the following examples (with bars shown as if we compute square root by division procedure).

Square root of numbers	No. of bars	No. of digits in the square root	Square root of numbers	No. of bars	No. of digits in the square root
$\sqrt{\overline{169}} = 13$	2	2	$\sqrt{\overline{4356}} = 66$	2	2
$\sqrt{\overline{441}} = 21$	2	2	$\sqrt{\overline{6084}} = 78$	2	2
$\sqrt{\overline{12544}} = 112$	3	3	$\sqrt{\overline{27225}} = 165$	3	3

Hence, we conclude that the number of bars indicates the number of digits in the square root.

Try these

Without calculating the square root, guess the number of digits in the square root of the following numbers:

1. 14400 2. 390625 3. 100000000

1.7.4 Square root of decimal numbers

To compute the square root of numbers in the decimal form, we simply follow the following steps:

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